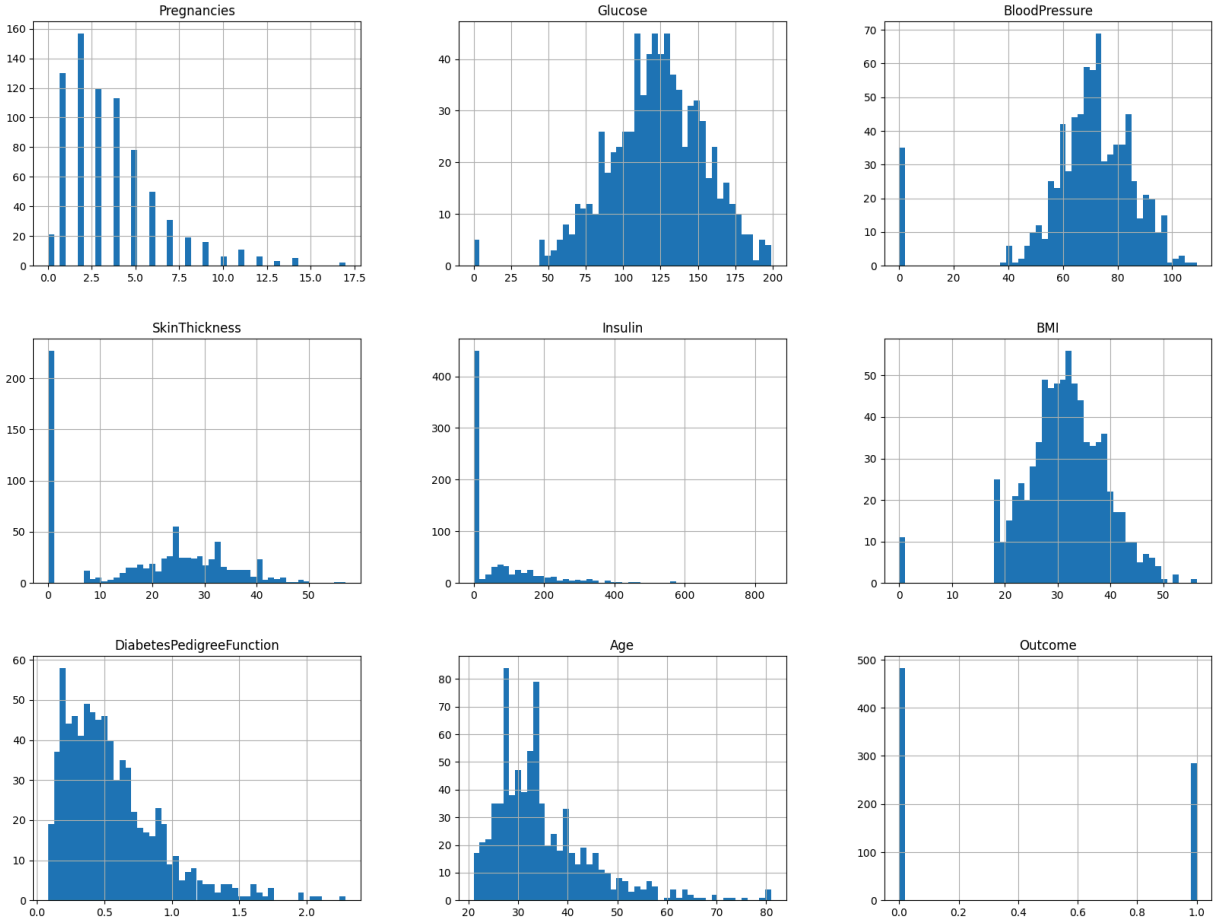
Experiment 5. Data collection and initial exploration

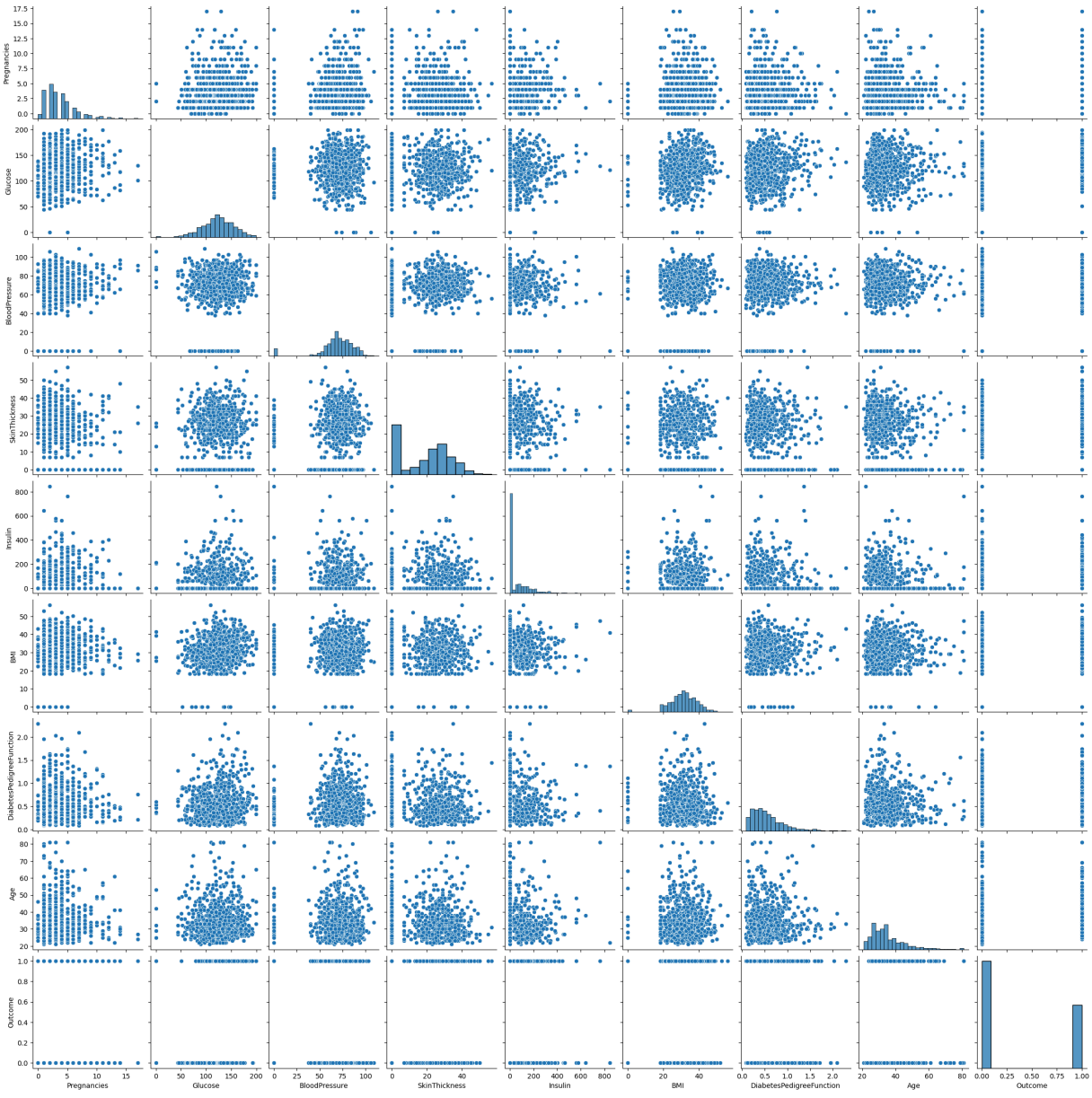
In [1]:
```
import pandas as pd
db= pd.read_csv("/content/PIMA Indian Dataset.csv")
print(db.head())
print(db.info())
print(db.describe())
import matplotlib.pyplot as plt
import seaborn as sns
db.hist(bins=50, figsize=(20, 15))
plt.show()
sns.pairplot(db)
plt.show()
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.717448  122.665365      69.063802      19.078125   67.562500
std       2.719788   31.928150      19.333074      14.426784  111.427159
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       2.000000  103.000000      63.000000       0.000000    0.000000
50%       3.000000  123.500000      71.000000      22.500000    0.000000
75%       5.000000  145.000000      81.000000      30.000000  106.250000
max      17.000000  199.000000     109.000000      57.000000  846.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    31.447526                  0.551858   34.798177    0.371094
std      7.815766                  0.347536    9.952149    0.483412
min      0.000000                  0.083000   21.000000    0.000000
25%     27.200000                  0.290000   28.000000    0.000000
50%     31.600000                  0.477000   32.000000    0.000000
75%     36.700000                  0.712000   39.000000    1.000000
max     56.300000                  2.288000   81.000000    1.000000
```


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome